CONOR DYSON

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Postdoctoral researcher in Clément Bonnerot's tidal disruption event group at the University of Birmingham. Research in black hole perturbation theory, relativistic astrophysics and self-force theory: post-adiabatic modelling of extreme and intermediate mass ratio inspirals, and the matter environments around black holes and how they shape the motion of bodies in strong-field gravity.

CURRENT POSITION

Postdoctoral Researcher, School of Physics and Astronomy and Institute for Gravitational Wave Astronomy, University of Birmingham — 2026–Present

RESEARCH INTERESTS

Black Holes, Gravitational Waves, General Relativity, Extreme Mass Ratio Inspirals, Environments, Self-Force, Dark Matter, and Relativistic Fluids

EDUCATION

PhD — **Niels Bohr Institute, University of Copenhagen**, *Motion Through Matter in Curved Space*, Supervisors: Maarten van de Meent *&* Vitor Cardoso — 2022–2026

Massachusetts Institute of Technology, Visiting Student (1 semester), Host: Scott Hughes — 2025

MASt — **DAMTP, University of Cambridge**, Applied Mathematics. *Perturbations of a Kerr Black Hole*, Essay supervisor: Harvey Reall — 2021–2022

BSc — **University College Dublin**, Applied and Computational Mathematics. *Ray Tracing and Neutron Star Visualisation*, Supervisor: Sarp Akcay — 2017–2021

University of Edinburgh, Erasmus Student (1 semester) — 2020

AWARDS & HONOURS

Best Scientific Poster, Black Holes Inside and Out, Strong gravity group. — 2024

Ad Astra Scholarship, University College Dublin — 2018-2021

BSc Stage 4 Scholarship, University College Dublin — 2020

Fr. Ciaran Ryan Prize, University College Dublin — 2020

PUBLICATION SUMMARY

8 publications (incl. 1 PRL), 5 as first author, 232 citations (h-index: 6)

INVITED SEMINARS

Environments Discussion Panel, LISA Waveform Working Group Meeting, **SISSA, Trieste** (expected) — Feb. 2026

Environments and Fundamental Physics Discussion Panel, Capra Meeting, **University of Southampton** — July 2025

Self-Force Seminar, **University of Southampton** — July 2025

Coffee Talk, **Institute for Advanced Study** — May 2025

Coffee Talk, **Centre for Theoretical Physics, MIT** — May 2025

ACR group meeting presentation, **Albert Einstein Institute** — Nov. 2023

CONTRIBUTED TALKS

24th International Conference on General Relativity, **Glasgow, Scotland**	July 2025
28th Capra Meeting on Radiation Reaction in GR, **Southampton, England**	July 2025
YITP long-term workshop in Gravity and Cosmology, **Kyoto, Japan**	Feb. 2024
26th Capra Meeting on Radiation Reaction in GR, **Copenhagen, Denmark**	June 2024

POSTER PRESENTATIONS

LISA Symposium, **Dublin, Ireland**	July 2025
Black Holes Inside and Out, **Copenhagen, Denmark**	Aug. 2025

ORGANISATION

Strong Group Weekly Meetings and Seminars, Copenhagen (Denmark)	2024–2026
Strong Group Retreat, Fuglsang Manor (Denmark)	June 2025
Niels Bohr International Academy Barbecue, Copenhagen (Denmark)	Sept. 2024
Post-linear Meetings, Niels Bohr Institute, Copenhagen (Denmark)	2024
Capra Conference on Radiation Reaction in General Relativity, Copenhagen (Denmark)	June 2023

INTERNATIONAL VISITS

Massachusetts Institute of Technology (3 months). *Host: Scott Hughes*	2025
Johns Hopkins University (1 week). *Host: Daniel D'Orazio*	2025
University of Southampton (1 week). *Host: Adam Pound*	2025
Institute for Advanced Study (2 days). *Host: Giovanni Tomaselli*	2025
Yukawa Institute for Theoretical Physics, Kyoto University (5 weeks)	2024
Albert Einstein Institute (1 week). *Host: Maarten van de Meent*	2023

COLLABORATIONS

Multi-Scale Self-Force Collaboration	2023–Present
LISA Consortium, Project Coordinator & Core Member	2025–Present
Black Hole Perturbation Toolkit, Contributor	2023–Present

TEACHING

Master student supervision: Manon Johansson (15 ECTS)	2025/26
Master student supervision: Ziying Ma (7.5 ECTS)	2025/26
TA: Stochastic Inverse Problems (Lecturer: K. Mosegaard)	2023/24
TA: Computational Science (Lecturer: J. Samsing)	2023

(Teaching at the University of Copenhagen)

SOFTWARE

Expert in Mathematica. Proficient in Python and MATLAB. Proficient with xAct and the Black Hole Perturbation Toolkit

SERVICE

LISA Consortium: Project Coordinator
Referee for **PRD**, **PRL**, **CQG**, and **SciPost**